

PRACTICE SET 2

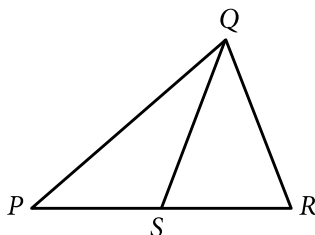
Quantitative Comparison Questions

For Questions 1 to 9, compare Quantity A and Quantity B, using additional information centered above the two quantities if such information is given. Select one of the following four answer choices and fill in the corresponding oval to the right of the question.

- (A) Quantity A is greater.
- (B) Quantity B is greater.
- (C) The two quantities are equal.
- (D) The relationship cannot be determined from the information given.

A symbol that appears more than once in a question has the same meaning throughout the question.

	Quantity A	Quantity B	Correct Answer
Example 1:	$(2)(6)$	$2 + 6$	(A) (B) (C) (D)



	Quantity A	Quantity B	Correct Answer
Example 2:	PS	SR	(A) (B) (C) (D) (since equal lengths cannot be assumed, even though PS and SR appear equal)

	$x < 0$		
	Quantity A	Quantity B	
1.	x^5	x^4	(A) (B) (C) (D)

- | | <u>Quantity A</u> | <u>Quantity B</u> | |
|----|-------------------|-------------------|-----------------|
| 2. | $(x + 4)(y + 3)$ | $(x + 3)(y + 4)$ | (A) (B) (C) (D) |
-

$0.\overline{b}$ represents the decimal in which the digit b is repeated without end.

- | | <u>Quantity A</u> | <u>Quantity B</u> | |
|----|-----------------------------------|-------------------|-----------------|
| 3. | $0.\overline{3} + 0.\overline{7}$ | 1.0 | (A) (B) (C) (D) |
-

A company plans to manufacture two types of hammers, type R and type S . The cost of manufacturing each hammer of type S is \$0.05 less than twice the cost of manufacturing each hammer of type R .

- | | <u>Quantity A</u> | <u>Quantity B</u> | |
|----|---|---|-----------------|
| 4. | The cost of manufacturing 1,000 hammers of type R and 1,000 hammers of type S | The cost of manufacturing 1,500 hammers of type S | (A) (B) (C) (D) |
-

- | | <u>Quantity A</u> | <u>Quantity B</u> | |
|----|---|-------------------|-----------------|
| 5. | $\frac{\left(\frac{1}{4}\right)^{-1} + \left(\frac{1}{4}\right)^{-2} + \left(\frac{1}{4}\right)^{-3}}{4}$ | 21 | (A) (B) (C) (D) |
-

List X : 2, 5, s , t

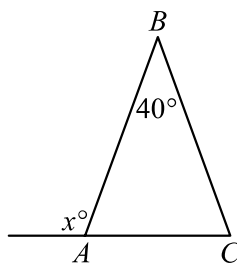
List Y : 2, 5, t

The average (arithmetic mean) of the numbers in list X is equal to the average of the numbers in list Y .

- | | <u>Quantity A</u> | <u>Quantity B</u> | |
|----|-------------------|-------------------|-----------------|
| 6. | s | 0 | (A) (B) (C) (D) |

The radius of circle A is 12 greater than the radius of circle B .

- | | <u>Quantity A</u> | <u>Quantity B</u> | |
|----|---|-------------------|-----------------|
| 7. | The circumference of circle A minus the circumference of circle B | 72 | (A) (B) (C) (D) |
-



$$AB = BC$$

- | | <u>Quantity A</u> | <u>Quantity B</u> | |
|----|-------------------|-------------------|-----------------|
| 8. | x | 120 | (A) (B) (C) (D) |
-

$$S = \{1, 2, 3, 4, 5, 6, 7, 8\}$$

- | | <u>Quantity A</u> | <u>Quantity B</u> | |
|----|---------------------------------------|---------------------------------------|-----------------|
| 9. | The number of 4-member subsets of S | The number of 5-member subsets of S | (A) (B) (C) (D) |

Multiple-choice Questions—Select One Answer Choice

For Questions 10 to 17, select a single answer choice.

10. In right triangle ABC , the ratio of the lengths of the two legs is 2 to 5. If the area of triangle ABC is 20, what is the length of the hypotenuse?
- (A) 7
(B) 10
(C) $4\sqrt{5}$
(D) $\sqrt{29}$
(E) $2\sqrt{29}$
11. According to surveys at a company, 20 percent of the employees owned cell phones in 1994, and 60 percent of the employees owned cell phones in 1998. From 1994 to 1998, what was the percent increase in the fraction of employees who owned cell phones?
- (A) 3%
(B) 20%
(C) 30%
(D) 200%
(E) 300%
12. If t is an integer and $8m = 16^t$, which of the following expresses m in terms of t ?
- (A) 2^t
(B) 2^{t-3}
(C) $2^{3(t-3)}$
(D) 2^{4t-3}
(E) $2^{4(t-3)}$
13. Three pumps, P , R , and T , working simultaneously at their respective constant rates, can fill a tank in 5 hours. Pumps P and R , working simultaneously at their respective constant rates, can fill the tank in 7 hours. How many hours will it take pump T , working alone at its constant rate, to fill the tank?
- (A) 1.7
(B) 10.0
(C) 15.0
(D) 17.5
(E) 30.0

14. The perimeter of a flat rectangular lawn is 42 meters. The width of the lawn is 75 percent of its length. What is the area of the lawn, in square meters?
- Ⓐ 40.5
Ⓑ 96
Ⓒ 108
Ⓓ 192
Ⓔ 432
15. The greatest of the 21 positive integers in a certain list is 16. The median of the 21 integers is 10. What is the least possible average (arithmetic mean) of the 21 integers?
- Ⓐ 4
Ⓑ 5
Ⓒ 6
Ⓓ 7
Ⓔ 8
16. If x and y are integers and $x = \frac{(2)(3)(4)(5)(7)(11)(13)}{39y}$, which of the following could be the value of y ?
- Ⓐ 15
Ⓑ 28
Ⓒ 38
Ⓓ 64
Ⓔ 143

17. Of the 40 specimens of bacteria in a dish, 3 specimens have a certain trait. If 5 specimens are to be selected from the dish at random and without replacement, which of the following represents the probability that only 1 of the 5 specimens selected will have the trait?

(A) $\frac{\binom{5}{1}}{\binom{40}{3}}$

(B) $\frac{\binom{5}{1}}{\binom{40}{5}}$

(C) $\frac{\binom{40}{3}}{\binom{40}{5}}$

(D) $\frac{\binom{3}{1}\binom{37}{4}}{\binom{40}{3}}$

(E) $\frac{\binom{3}{1}\binom{37}{4}}{\binom{40}{5}}$

Multiple-choice Questions—Select One or More Answer Choices**For Questions 18 to 19, select all the answer choices that apply.**

18. Two different positive integers x and y are selected from the odd integers that are less than 10. If $z = x + y$ and z is less than 10, which of the following integers could be the sum of x , y , and z ?

Indicate all such integers.

- ☐ A 8
- ☐ B 9
- ☐ C 10
- ☐ D 12
- ☐ E 14
- ☐ F 15
- ☐ G 16
- ☐ H 18

19. For a certain probability experiment, the probability that event A will occur is $\frac{1}{2}$ and the probability that event B will occur is $\frac{1}{3}$. Which of the following values could be the probability that the event $A \cup B$ (that is, the event A or B , or both) will occur?

Indicate all such values.

- ☐ A $\frac{1}{3}$
- ☐ B $\frac{1}{2}$
- ☐ C $\frac{3}{4}$

Numeric Entry Questions

For Questions 20 to 21, enter your answer in the answer box(es) below the question.

- Your answer may be an integer, a decimal, or a fraction, and it may be negative.
- If a question asks for a fraction, there will be two boxes—one for the numerator and one for the denominator. A decimal point cannot be used in a fraction.
- Equivalent forms of the correct answer, such as 2.5 and 2.50, are all correct. Fractions do not need to be reduced to lowest terms, though you may need to reduce your fraction to fit in the boxes.
- Enter the exact answer unless the question asks you to round your answer.

20. If a and b are the two solutions of the equation $x^2 - 5x + 4 = 0$, what is the value of $\left(\frac{1+a}{a}\right)\left(\frac{1+b}{b}\right)$?

Give your answer as a fraction.

21. From 2011 to 2012, Jack's annual salary increased by 10 percent and Arnie's annual salary decreased by 5 percent. If their annual salaries were equal in 2012, then Arnie's annual salary in 2011 was what percent greater than Jack's annual salary in 2011?

Give your answer to the nearest 0.1 percent.

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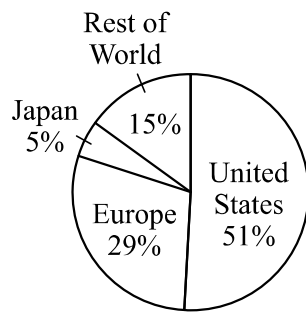
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Data Interpretation Set

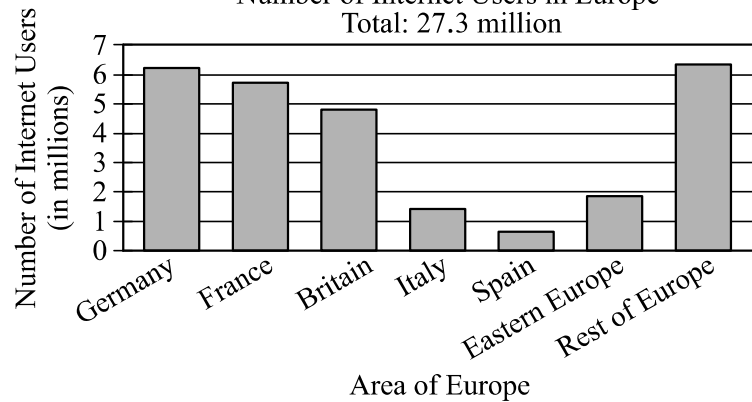
Questions 22 to 25 are based on the following data. For these questions, select a single answer choice unless otherwise directed.

INTERNET USE IN YEAR X

Distribution of Internet Users
Worldwide, by Region



Number of Internet Users in Europe
Total: 27.3 million



22. Which of the following is closest to the percent of Internet users in Europe who were in countries other than Germany, France, Britain, Italy, and Spain?
 - (A) 30%
 - (B) 34%
 - (C) 38%
 - (D) 42%
 - (E) 46%
23. Approximately what was the range of the numbers of Internet users in the seven areas of Europe shown in the bar graph?
 - (A) 6.5 million
 - (B) 5.5 million
 - (C) 3.5 million
 - (D) 3.0 million
 - (E) 2.5 million
24. The number of Internet users in the United States was approximately how many times the number of Internet users in Italy?
 - (A) 5
 - (B) 15
 - (C) 20
 - (D) 25
 - (E) 35

For Question 25, select all the answer choices that apply.

25. Based on the information given, which of the following statements about Internet use in year X must be true?

Indicate all such statements.

- ☐ A The United States had more Internet users than all other countries in the world combined.
- ☐ B Spain had fewer Internet users than any country in Eastern Europe.
- ☐ C Germany and France combined had more than $\frac{1}{3}$ of the Internet users in Europe.